

PH113 / Tutorial Questions - Module II: EM Theory

(Dated: February 2, 2024)

1. Gradient, Divergence, and Curl.

- (a) Calculate the gradient of (i) $U = x^2$, (ii) $U = r^2$, (iii) $\mathbf{c} \cdot \mathbf{r}$ ($\mathbf{c}$ is a constant vector).
(b) For $U = f(r)$, show that $\nabla U = \frac{df}{dr} \left[\frac{\mathbf{r}}{r} \right]$.

*Comment: ∇U has the property that the rate of change of U with respect to (w.r.t) distance in a particular direction ($\mathbf{d}$) is the projection of ∇U onto that direction (or the component of ∇U in that direction. Mathematically, $dU/ds = \nabla U \cdot d\mathbf{r}/ds$, where dU/ds is the **directional derivative** of U . Of course, it does not make sense to talk about directional derivatives without specifying the direction.*

We could also say that, at any point P , ∇U points in the direction of greatest change of U at P , and the magnitude equals the rate of change of U with distance in that direction.

Also, ∇U is normal to a surface of constant U .

- (c) Calculate the divergence of (i) $x\hat{\mathbf{i}}$, (ii) $\mathbf{r}$, (iii) $\mathbf{r}/r^3$, (iv) $r\mathbf{c}$ ($\mathbf{c}$ is a constant).

Comment: The divergence of a vector field represents the flux generation per unit volume at each point of the field. (Divergence because it is an efflux not an influx.)

- (d) Calculate the curl of (i) $-y\hat{\mathbf{i}} + x\hat{\mathbf{j}}$, (ii) $x^2y^2\hat{\mathbf{k}}$.

*Comment: The curl of the vector field represents the **vorticity**, or **circulation per unit area** of the vector field.*

- (e) Show that $\nabla \cdot (\nabla \times \mathbf{v}) = 0$
(f) Show that $\nabla \times (\nabla f) = 0$

*Comment: Remember, "d c g(enerator)", which stands for **d**iv of a **c**url is zero, **c**url of a **g**rad is zero!*

Comment:

- A vector field with zero divergence is said to be **solenoidal**.
- A vector field with zero curl is said to be **irrotational**.
- A scalar field with zero gradient is said to be, er, well, **constant**.

- (g) Show that the curl of a $\vec{r}$ is zero. In general, derive the conditions when the curl of a radial vector field is zero.

*Comment: This is useful to prove that a **static** electric field is conservative. A common mistake that students make is to already introduce the notion of scalar potential as $\mathbf{E} = -\nabla V$ (and not as a work done per unit charge !). **There is a fundamental problem with this approach.** More on this, at the end.*

- (h) Calculate the Laplacian of (i) r^2 , (ii) xy^2z^3 , (iii) $1/r$.

Comment: Laplacian is given by $\nabla \cdot (\nabla U) = \nabla^2 U$, where U is a scalar field.

- (i) Explicit statements of Divergence and Stoke's theorems (mathematical forms).

2. Gauss's law.

- (a) Use Gauss's law to find the electric field inside a uniformly charged sphere (charge density ρ) of radius R .
- (b) Find the electric field inside a sphere which carries a charge density proportional to the distance from the origin: $r = kr$, for some constant k .
- (c) Electric field due to an infinite charge sheet with charge density σ .
- (d) [!] If the electric field in some region is given by:

$$\mathbf{E} = \frac{A\hat{r} + B \sin \theta \cos \phi \hat{r}}{r},$$

where A and B are constants, what is the charge density?

Comment: Hint: Use Gauss's law in spherical polar coordinates.

- (e) For a point charge of 1×10^{-6} C at the centre of a cubical Gaussian surface of sides 0.5 m, determine the flux from the surface?

- (f) [!] Suppose the electric potential is given by the expression $V(r) = Ae^{-kr}/r$, for all r (A and k are constants). Find the electric field $\mathbf{E}(r)$, the charge density $\rho(r)$, and the total charge Q .

Comment: This is called the Yukawa potential, or the screened Coulomb potential, and find numerous applications in the current day research in condensed matter physics as well as in nuclear and particle physics.

This is also an example of a regularizing function: For a badly behaved function, such as $1/r$ which has a singularity (blows up!) at $r = 0$, it can be multiplied by another function which “controls” the rate of approaching the singularity. In the above case, the exponential function approaches 1 much faster than the rate at which $1/r$ blows up. Therefore, the e^{-kr} is a regularizing function. All calculations can now be done, and in the end we take the limit $k \rightarrow 0$.

Comment: In order to solve this problem, you will need $\nabla \cdot (\hat{r}/r^2) = \delta(\mathbf{r})$. The delta function is called the Dirac delta function and means that the value of this continuous function is zero everywhere except at $\mathbf{r} = \mathbf{0}$. In general:

$$\delta(x - a) = 1 \text{ if } x = a; 0 \text{ otherwise.}$$

3. Determination of $\mathbf{E}$ for some distribution of charges. For example,

- (a) Consider two charges q_1 and q_2 at $\pm d/2$ from the origin, along the x -axis. Obtain the electric field at (i) distance z away from the origin along z , (ii) along the distance $x < -d/2$ and $x > +d/2$ along the x axis.
- (b) $+q$ and $-q$ charges in the same geometry.

Comment: This is the concept of an electric dipole. Note what happens for situations similar to (ii) in the problem above.

- (c) What happens is the charge q is smeared over length L along the x -axis centered at origin?
- (d) $\mathbf{E}$ at some point along the axis of symmetry for a uniform charge distributed as a ring.

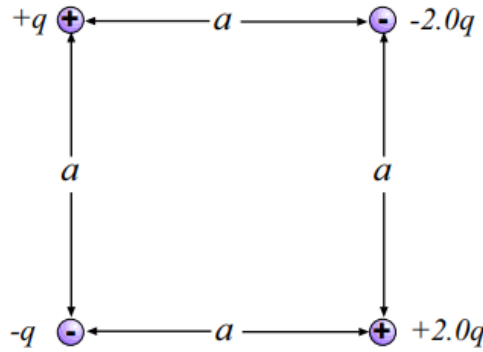


FIG. 1. Charge distribution for Q3(e).

| *Comment: Similar questions for computation of electric potential.*

- (e) Calculate the electric field at a distance z away along the axis of a disk with uniform charge density σ . What happens when the radius of the disk tends to be infinitely large?
4. Consider the situation shown in the figure below. Calculate the electric field at the center of the square if $q = 1.0 \times 10^{-8}$ C and $a = 5.0$ cm.
5. An object having a net charge of $24 \mu\text{C}$ is placed in a uniform electric field of 610 N/C directed vertically. What is the mass of this object if it “floats” in the field?
6. Show that $\mathbf{E} = k[(y^2)\hat{i} + (2yz + z^2)\hat{j} + (2yz)\hat{k}]$, where k is a constant, is an impossible electrostatic field.
7. Show that $\mathbf{E} = k[(xy)\hat{i} + (2yz)\hat{j} + (3xz)\hat{k}]$ is a valid electrostatic field, and obtain the potential using origin as the reference point. Check your answer by computing ∇V .
8. [!] An electric quadrupole consists of two dipole moments of equal magnitude but in opposite direction. Show that the value of E on the axis of the quadrupole for points at distance $z \gg d$ away from its center is

$$E = \frac{3Q}{4\pi\epsilon_0 z^4},$$

where $Q = 2qd^2$ is **the quadrupole moment** of the charge distribution.

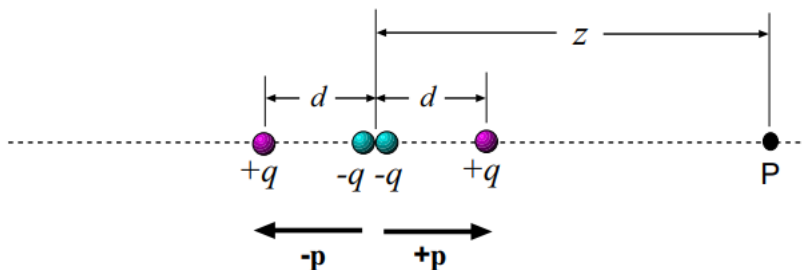


FIG. 2. Re: Q 4; The charge configuration of an electric quadrupole.

9. A circular current loop of radius R carrying a current I is placed in the xy -plane. A constant uniform magnetic field cuts through the loop parallel to the y -axis. Find the magnetic force on the upper half of the loop, the lower half of the loop, and the total force on the loop.

*Comment: **Significance.** The total force on any closed loop in a uniform magnetic field is zero. Even though each piece of the loop has a force acting on it, the net force on the system is zero. (Note that there will be a net torque on the loop.)*

10. What is magnetic field produced by an infinite wire carrying current I at a distance d from the wire.
11. Use the above to calculate the magnitude of the magnetic field produced by a straight current-carrying conductor at a distance of 2 m from it when the current through the conductor is 10A. What about the direction of B .
12. A straight current-carrying conductor is carrying a current of 10 A and another conductor parallel to it carries a current of 5 A on the opposite side as shown in the figure below. Find the magnitude of the magnetic field produced by the system at a distance of 2 m.
13. Write down the static and time-dependent Maxwell's equation in vacuum.

Comment: In principle, one should be able to write down the equations in differential as well as integral form. All in all, there are 4×2 equations for vacuum and 4×2 equations in media. You may also be asked for the "source-free form" in which case you should set the charge density and the current density to zero; $\rho = 0$, $\mathbf{J} = \mathbf{0}$.

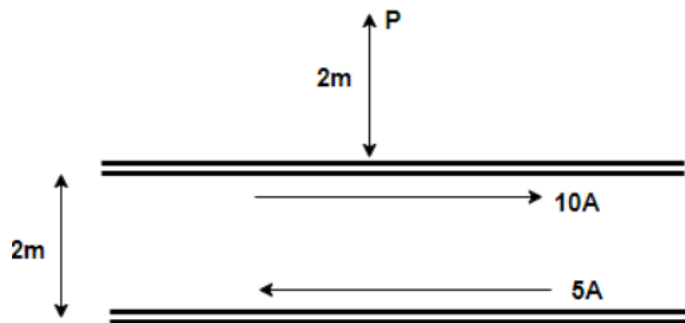


FIG. 3. For Q10

14. [!] Show that a charge particle moving perpendicular to an external magnetic field undergoes a circular motion with angular frequency $\omega = qB/m$, where q , B and m are the charge on the particle, the strength of the magnetic field, and mass of the particle.

Comment: What would happen in the initial velocity of particle is not perpendicular to $\mathbf{B}$?

*Based on the above problem, the component of the velocity perpendicular to $\mathbf{B}$ will induce a circular motion, while the component along $\mathbf{B}$ will translate the charged particle, leading to a **helical motion** (see Fig. 4).*

15. Explicitly derive the boundary conditions for $\mathbf{E}$ and $\mathbf{B}$ at an interface between two media.
16. The square coil with sides $l = 0.25$ m long and is tightly wound with $N = 200$ turns of wire. The resistance of the coil is $R = 5.0 \, \Omega$. The coil is placed in a spatially uniform magnetic field that is directed perpendicular to the face of the coil and whose magnitude is decreasing at a rate $dB/dt = -0.040$ T/s. (a) What is the magnitude of the emf induced in the coil? (b) What is the magnitude of the current circulating through the coil?

Comment: If the area of the loop were changing in time, we would not be able to pull it out of the time derivative. Since the loop is a closed path, the result of this current would be a small amount of heating of the wires until the magnetic field stops changing. This may increase the area of the loop slightly as the wires are heated.

17. Using the capacitor geometry set-up and/or the conservation of charges, argue the need for displacement current.

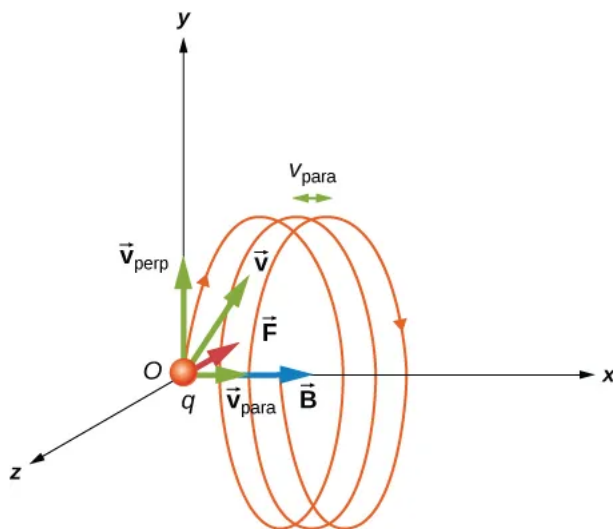


FIG. 4. A charged particle moving with a velocity not in the same direction as the magnetic field. The velocity component perpendicular to the magnetic field creates circular motion, whereas the component of the velocity parallel to the field moves the particle along a straight line. The pitch is the horizontal distance between two consecutive circles. The resulting motion is helical.

18. Statement and meaning of charge continuity equation.
19. Explicitly write down the time-dependent Maxwell's equations (in vacuum as well as in media). **Also learn the names of these equation, e.g. Gauss's Law, Ampere's (circuital) law, etc.**
20. Meaning of $\nabla \cdot \mathbf{B} = 0$
21. Write the expression for Lorentz force (involving both E and B).
22. Meaning of D and H; relation between ϵ and ϵ_0 , μ and μ_0 .

Comment: It is important to note that although $\mathbf{D} = \epsilon_0 \mathbf{E}$ is correct as far as changing variables is concerned, it contains no physical meaning. It is particularly problematic to use this form of $\mathbf{D}$ since it is generally assumed that $\mathbf{D}$ is the effective electric field in a dielectric medium, and the above holds if and only if the induced polarization is identically zero.

Similarly, $\mathbf{B}$ and $\mathbf{H}$ are related through the magnetization of the medium.

*Comment: **General remark.** The relation between electric field and electrostatic potential $\mathbf{E} = -\nabla V$ is possible since the electrostatic field is conservative, implying $\nabla \times \mathbf{E} = \mathbf{0}$. Using the fact that curl of a gradient is **always** zero, implying that $\mathbf{E}$ can be written as proportional to a gradient term. The sign in front of gradient is due to the fact that the reference in electrostatics is at ∞ . In other words, $V(\infty) = 0$.*

To prove the conservative-ness of electrostatic $\mathbf{E}$, it would suffice to show that curl of $\vec{r}/r^2$ is zero. It is, in fact, wrong to assume that a $V(r)$ already exists and its relation with $\mathbf{E}$ is known.

*Nevertheless, $V(r)$ can, in principle, be defined as **the work done per unit charge** and one may then proceed to show that $V(r)$ are related via a gradient.*